

Price watch

Are you on the lookout for a set of speakers to serve your aural needs? Here's what you need

**Space: The Final Frontier**

What will future spaceships look like? Here's what some scientists think:
<http://dgit.in/1t4NX08>

Do you need an AC?

We're talking about routers and not air conditioners. The latest Wi-Fi protocol revision offers a lot of benefits over the previous revision but does the average guy need them?

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Does it have AC Wi-Fi? Not a question one hears normally, not even when we're talking about phones. We get plenty of emails regarding routers and very few people are actually aware of this Wi-Fi protocol. Most of the queries are centered on compatibility with existing devices that people have. Then again, if you are going to invest in new hardware it better be worth the money right?. This brings to mind the adoption of the product. To put things into perspective let's take phones into consideration. At the time of writing this I've found 73 phones that are capable of Wi-Fi 802.11ac compared to 1577 phones that support 802.11n. Analyst predictions were around 8%

for the year 2014 while actual figures are roughly half that number. There are many laptops, phones, tablets, etc. that supports 802.11ac but most of these devices are from the premium segment, and even in that segment, it's hard to find support for ac (case in point, the iPad Air)

Moreover, we don't have services Netflix with all its glorious HD content that we'd be able to enjoy without ever seeing the "Buffering circle". So in the end, you don't have a lot of bandwidth for online content. When it comes to streaming locally, you're not really going to need the extra bandwidth that ac offers unless two or more people are streaming HD content at the same time.

Even in such a scenario, the element of jugaad always exists. Not everyone has a penchant for viewing only uncompressed

full HD content and even if you do, the chances that there are two people connected to the same network that'd be streaming at the exact same time is miniscule. But let's say all of the above applies to your scenario, I'd say you're one of the very few people for whom a 802.11ac device is necessitated and by all means do buy a 802.11ac router with 1750-1900 Mbps bandwidth. The proponents for the tech-

nology claim an adoption of 45% by next year, a figure that we won't be seeing for at least two more years in reality.

If all of that wasn't convincing enough, then let's pull out the raw numbers.

Other than LAN data transfers, the only bandwidth intensive operation happens to be uncompressed HDMI video which needs twice the bandwidth that most 802.11ac devices can provide. Moreover, the people who can discern such a video stream from a slightly compressed stream of the same file is also a minority.

Enough about bandwidth, let's look at the other aspects of 802.11ac.

- 80 MHz channel bandwidth - More bandwidth
- MU-MIMO - Simultaneous connections are better serviced
- 256 QAM - More bandwidth
- Beam-forming - Better connectivity

Each of the above features are great but there isn't an urgent need for the same. It's not as if not having an 802.11ac is going to leave you bereft of an incredible experience or even cause the mildest inconvenience. All I'd say is, if your current router conked off and you had to buy a new one, then an 802.11ac router might be logical upgrade, albeit not a necessary one. But if your current router is working fine and you've bought a few 802.11ac enabled hardware, then you need not unnecessarily upgrade your router to get the "best experience". 

Application	Recommended bit rate
VoIP - voice transport	27 - 93 kbps (codec dependent)
VoIP - signaling (typically SIP)	5 kbps
Remote display	150 kbps (without video), 1.8 Mbps (with video)
Web conferencing	384 kbps - 1 Mbps
FaceTime	0.9 Mbps
AppleTV video streaming	2.5 - 8 Mbps
High-definition video (compressed)	2 - 5 Mbps
High-definition video (uncompressed)	20 Mbps
High-definition video (uncompressed HDMI)	3.3 Gbps
Standard-definition video	1 - 1.5 Mbps
Email/web browsing	0.5 - 1.0 Mbps
File sharing	5 Mbps
YouTube	0.9 Mbps
Network backup	More is better